

Marios Papachristou

Assistant Professor of Information Systems
W. P. Carey School of Business
Arizona State University

LAST UPDATED	September 27, 2026
PERSONAL INFORMATION	E-mail mpapachr@asu.edu Website papachristoumarios.github.io Google Scholar Profile Google Scholar Profile Campus Address 400 E Lemon St, BAC, Tempe, AZ 85281 US Status US Permanent Resident (Green Card Holder)
RESEARCH	My research develops models for understanding and improving the <i>resilience</i> of modern socio-technical systems with downstream applications to platforms, supply chains, and organizations. I study the mechanisms causing fragility and devise analytical metrics to measure the resilience, prescriptive interventions, and decentralized decision-making under uncertainty and information constraints.
METHODS	Network Science, Algorithms, AI Agents, Privacy, Computational Social Science, ML
ACADEMIC APPOINTMENTS	W. P. Carey School of Business, Arizona State University 2025 – Tenure-Track Assistant Professor, Department of Information Systems • <i>Affiliated Member</i>, Center for AI and Data Analytics (AIDA) • <i>Graduate Faculty Member</i>, Data Science, Analytics and Engineering • <i>Graduate Faculty Member</i>, Computer Science Cornell University Summer 2026 Visiting Faculty
EDUCATION	Cornell University 2020 – 2025 Ph.D. in Computer Science, Minor: <i>Applied Mathematics</i> – <i>Doctoral Advisor</i>: Jon Kleinberg – <i>Thesis Title</i>: “From Contagion to Stability: Insights into Network Dynamics, Resilience and Stability”. – Nominated for the <i>ACM Doctoral Dissertation Award</i>. M.S. in Computer Science 2020 – 2022 National Technical University of Athens 2015 – 2020 Diploma in Electrical and Computer Engineering – <i>Advisor</i>: Dimitris Fotakis
RESEARCH HIGHLIGHTS	$\alpha\beta$ = alphabetical order * = equal contribution ° = corresponding author ‡ = graduate student supervised [H1] Network Formation and Dynamics among Multi-LLMs <i>Marios Papachristou</i> , Yuan Yuan <i>PNAS Nexus</i> , 2025 [paper] [preprint] [code] – Taught at UC Berkeley, Stanford, UIUC and others – Featured by <i>TechXplore</i>, and <i>SingularityHub</i> [H2] The Power of Choice in Random Sampling and Applications to Platform Analytics <i>Marios Papachristou</i> , Jon Kleinberg

	<i>Major revision, Operations Research</i>, 2025 <pre>[preprint]</pre> – Taught at the University of Pittsburgh	
	[H3] Structural Measures of Resilience for Supply Chains <i>Marios Papachristou</i>, Amin Rahimian, Arash Azadegan <i>Major revision, Production and Operations Management</i>, 2025 <pre>[preprint]</pre> <code>[code]</code>	
	[H4] The Price of Misreporting: Measuring the Effect of Strategic Behavior in Online Platforms <i>Marios Papachristou</i>, Akhil Jalan Submitted, 2026 Preliminary version presented at the ACM Web Conference 2026 <pre>[preprint]</pre>	
	[H5] Quantifying Consensus Disparity in Social and Information Networks <i>Marios Papachristou</i>, Jon Kleinberg <i>Major revision, INFORMS Journal of Computing</i>, 2025 <pre>[preprint]</pre> <code>[code]</code>	
	[H6] Data Regulation and the Microstructure of Clinical Trials Kyuyeon (Jenny) Rhee[‡], <i>Marios Papachristou</i>, Tian Lu, Pei-yu Chen Under preparation for submission, 2026 Preprint available upon request – <i>INFORMS eBusiness Best Paper Award Finalist (best paper or runner-up)</i>	
INDUSTRY EMPLOYMENT	Microsoft Research Office of Applied Research, with Chin-Chia Hsu and Longqi Yang Twitter User Modeling Research Group, with Rishab Goel, Frank Portman, Matt Miller	Summer 2023 Summer 2022
HONORS & AWARDS	– INFORMS eBusiness Section Best Student Paper Award Finalist (best paper or runner-up). 2026 – ACM Doctoral Dissertation Award Nominee Nominated to the ACM by Cornell’s CS department as one of the two best Ph.D. theses in engineering-related disciplines awarded in 2025. 2025 – Sigma Xi Elected member based on scientific contributions. 2025 – Cornell ORIE Young Researchers Workshop Invited workshop. 2024 – Alexander A. Onassis Scholarship Three-year fellowship for Greek Ph.D. students. Acceptance rate: 16%. 2023 – Bodossaki Foundation Fellowship Runner-up. 2023 – LinkedIn Ph.D. Fellowship Awarded one-year fully funded fellowship by LinkedIn. Acceptance rate 14%. 2022 – A.G. Leventis Scholarship Award for graduate students of Greek origin. Acceptance rate 12.5%. 2022 – Gerondelis Scholarship Awarded one-time award for graduate students of Greek descent. 2022 – Chateaubriand Fellowship Awarded fellowship granted by the French Embassy in the US for studies in France (declined). Acceptance rate 33%. 2022 – Cornell Fellowship Awarded one-year fully funded fellowship by the CS department based on “exceptional preparation and promise”. 2020	

- **Thomaidion Award** 2019
Awarded by NTUA for publication during undergraduate studies.
- **ESEC/FSE 2019 ACM Student Research Competition Finalist** 2019
- **CanSat International Space Engineering Competition** 2019
Awarded 4th out of 93 teams. Organized by the American Astronautical Society (AAS).
- **2nd prize “be finnovative 2.0” Start-up Accelerator by National Bank of Greece** 2018
Awarded 2nd prize for our cashierless payments start-up.
- **1st prize at “Crowdhackathon Fintech #2” by the National Bank of Greece** 2017
Awarded 1st prize for our cashierless payments start-up.
- **IEEEExtreme 11.0 Programming Competition** 2017
Awarded top 1% worldwide in IEEE’s algorithms competition.
- **Touramanoglu Scholarship** 2015
Awarded for the highest University Entrance Exams score in the Imittos region.
- **“The Great Moment of Education” Scholarship by Eurobank SA** 2015
Awarded for the highest University Entrance Exams score in the Imittos region.

PUBLISHED PAPERS

Journal Papers

- [J1] **Network Formation and Dynamics among Multi-LLMs**
Marios Papachristou, Yuan Yuan
PNAS Nexus, 2025
[\[paper\]](#) [\[preprint\]](#) [\[code\]](#)
 - Taught at UC Berkeley, Stanford, UIUC and others
 - Featured by *TechXplore*, and *SingularityHub*
- [J2] **Leveraging Large Language Models for Collective Decision-Making**
Marios Papachristou, Longqi Yang, Chin-Chia Hsu
Proceedings of the ACM on Human Computer Interaction (presented at ACM CSCW ’25), 2025
[\[paper\]](#)
 - Invited talk at the LLMs as Research Tools Workshop at ACM CHI 2024
 - Featured by the *ACM*
- [J3] **volesti: A C++ library for sampling and volume computation on convex bodies**
 ^{$\alpha\beta$} Apostolos Chalkis*, Vissarion Fisikopoulos*, *Marios Papachristou*°, Elias Tsigaridas*
Journal of Open Source Software, 2025
[\[paper\]](#) [\[code\]](#)
- [J4] **Differentially Private Distributed Estimation and Learning**
Marios Papachristou, Amin Rahimian
IIE Transactions (formerly IIE Transactions), 2025
[\[paper\]](#) [\[code\]](#) [\[talk\]](#)
 - Featured at the *July 2025 issue of the ISE Magazine* (10% of articles published in IIE Transactions)
 - Featured by the *University of Pittsburgh Swanson School of Engineering*
- [J5] **Truncated Log-concave Sampling for Convex Bodies with Reflective Hamiltonian Monte Carlo**
 ^{$\alpha\beta$} Apostolos Chalkis, Vissarion Fisikopoulos, *Marios Papachristou*°, Elias Tsigaridas
ACM Transactions on Mathematical Software (TOMS), 2023
[\[paper\]](#) [\[code\]](#)
 - Featured at *ODBMS.org*
- [J6] **Sublinear Domination and Core-periphery Networks**
Marios Papachristou (solo-authored)
Scientific Reports, 2021
[\[paper\]](#) [\[code\]](#)
- [J7] **Commons-based peer production and digital fabrication: The case of a RepRap-based, Lego-built 3D printing-milling machine**
Vasilis Kostakis*, and *Marios Papachristou**

Telematics and Informatics, 2014
[paper]

Top-tier CS Conference Proceedings

- [C1] **Opinion Dynamics with Multiple Adversaries**
Akhil Jalan*, *Marios Papachristou**^o
WWW '26: *ACM Web Conference (WWW)*, 2026
[paper]
– Acceptance rate: 20%
- [C2] **Dynamic Interventions for Networked Contagions**
Marios Papachristou, Sid Banerjee, Jon Kleinberg
WWW '23: *ACM Web Conference (WWW)*, 2023
[paper] [code]
– Acceptance rate: 17%
- [C3] **Core-periphery Models for Hypergraphs**
Marios Papachristou, Jon Kleinberg
KDD '22: *ACM Intl. Conf. on Knowledge Discovery and Data Mining (KDD)*, 2022
[paper] [code] [data]
– Acceptance rate: 15%
– Taught at tutorials at KDD and CIKM
- [C4] **Allocating Stimulus Checks in Times of Crisis**
Marios Papachristou, Jon Kleinberg
WWW '22: *ACM Web Conference (WWW)*, 2022
[paper] [code]
– Acceptance rate: 17%
– Featured by the *Federal Reserve Bank of Minneapolis*
– Taught at UC Berkeley
- [C5] **Software Clusterings with Vector Semantics and the Call Graph**
Marios Papachristou (solo-authored)
ESEC/FSE '19: *ACM Joint Meeting on European Software Engineering Conference and Symposium on the Foundations of Software Engineering (ESEC/FSE)*, 2019
[paper] [code] [data]
– Acceptance rate: 21%
– ESEC/FSE Student Research Competition Finalist Paper

REVISIONS

- [R1] **The Power of Choice in Random Sampling and Applications to Platform Analytics**
Marios Papachristou, Jon Kleinberg
Major revision, Operations Research, 2025
[preprint]
– Taught at the University of Pittsburgh
- [R2] **Structural Measures of Resilience for Supply Chains**
Marios Papachristou, Amin Rahimian, Arash Azadegan
Major revision, Production and Operations Management, 2025
[preprint] [code]
- [R3] **Quantifying Consensus Disparity in Social and Information Networks**
Marios Papachristou, Jon Kleinberg
Major revision, INFORMS Journal of Computing, 2025
[preprint] [code]

WORKING PAPERS

- [WP1] **Data Regulation and the Microstructure of Clinical Trials**
Kyuyeon (Jenny) Rhee[‡], *Marios Papachristou*, Tian Lu, Pei-yu Chen

Under preparation for submission, 2026

Preprint available upon request

– **INFORMS eBusiness Best Paper Award Finalist (best paper or runner-up)**

- [WP2] **Differentially Private Distributed Inference for Multicenter Clinical Studies**
Marios Papachristou, Amin Rahimian
Submitted, 2026
Preliminary version presented at the Theory and Practice of Differential Privacy Workshop (TPDP)
[preprint]
- [WP3] **The Price of Misreporting: Measuring the Effect of Strategic Behavior in Online Platforms**
Marios Papachristou, Akhil Jalan
Submitted, 2026
Preliminary version presented at the ACM Web Conference 2026
[preprint]
- [WP4] **Optimal Resource Allocation for Remediating Networked Contagions**
Marios Papachristou, Sid Banerjee, Jon Kleinberg
Submitted, 2026
[preprint]
- [WP5] **From Thick Description To Computational Simulation: An Agentic Framework For Deriving Simulation Studies From Qualitative Research**
Julian Lehmann, David Dobolyi, *Marios Papachristou*, Gerald Kane
Submitted, 2026
Preprint available upon request

WORKS

IN PROGRESS

- [WIP1] **Impact of Massive AI-generated Content on Recommender Systems**
with Joonhwa Park[‡], and Panos Adamopoulos
- [WIP2] **The Effects of Generative AI on Consumer Behavior**
with Panos Adamopoulos

PUBLISHED SOFTWARE

volesti github.com/GeomScale/volesti
Open-source C++ statistical software for sampling and Bayesian inference.

REFEREED CS WORKSHOPS

- [W1] **GLINKX: A Scalable Unified Framework for Homophilous and Heterophilous Graphs**
Marios Papachristou, Rishab Goel, Frank Portman, Matt Miller, Rong Jin
NeurIPS workshop on Graph Learning Frontiers (GLFrontriers), 2022
[paper] [poster]

TUTORIALS

- [T1] **Differential Privacy for Strategic Information Sharing and Learning: Foundations, Mechanisms, and Applications**
Amin Rahimian, Juba Ziani, *Marios Papachristou*, Yuxin Liu
Conference on Web and Internet Economics (WINE), 2025

TEACHING

Arizona State University, Instructor *Spring 2026*
CIS 509: Analytics for Unstructured Data

– Evaluation (out of 7.0, MSBA/MSBA/MSAIB): **6.8/7.0, 6.6/7.0, 6.4/7.0 (Spring 2026)**, Department Average: 6.2/7.0

Cornell University, Teaching Assistant *Spring 2023, Fall 2023*
CS 6850: The Structure of Information Networks (Ph.D.-level)

- Evaluation (out of 5.0): **5.0/5.0 (Spring 2023), 4.8/5.0 (Fall 2023)**

GRANTS

- **ALCF Director’s Discretionary Allocation Award (DD Award)** 2026
Argonne Leadership Computing Facility. Project: Large-scale LLM-based Simulations for Algorithmic Auditing of Recommendation Systems

MENTEES

- Kyuyeon (Jenny) Rhee (Ph.D. student at ASU)
- Joonhwa Park (Ph.D. student at Emory)
- Victor Lee (Ph.D. student at ASU)
- Pin-Jun Liu (Ph.D. student at ASU)
- Jasmine Li (Undergraduate Cornell CS)
- Ioannis Iakovidis (Google Summer of Code 2022; now Ph.D. student at Johns Hopkins)
- Zhang Zhuyan (Google Summer of Code 2022; now Technology analyst at Bank of America)

MENTORSHIP

EXPERIENCE

- Student-applicant Support Program at Cornell CIS 2022
- Google Summer of Code Mentor 2019 – 2022
- Volunteer lectures for the Greek National Olympiad of Informatics (PDP) 2019

INVITED TALKS &

LECTURES

† = scheduled

* = presented by co-author

- ***The Price of Misreporting: Measuring the Effect of Strategic Behavior in Online Platforms***
 - Eller College of Management, University of Arizona[†] November 2026
 - Lerner-Warwick FinTech Workshop[†] November 2026
 - NYU Stern School of Business, WAICI Workshop[†] September 2026
- ***Resilience in Networks: Frameworks for Resource Allocation and Simulation***
 - Archimedes, Athena RC April 2025
 - TOC4Fairness Seminar, Simons Institute March 2025
 - Cornell University, Theory Seminar December 2024
 - Arizona State University, W. P. Carey School of Business November 2024
 - Purdue University, Daniels School of Business November 2024
 - MIT, Research Group Meeting June 2024
- ***Network Formation and Dynamics among Multi-LLMs***
 - Arizona State University, School of Mathematical and Natural Sciences, AI Seminar November 2025
 - Arizona State University, Machine Learning Day April 2025
 - IBM Research, Exploratory Math Sciences Council Seminar April 2024
 - UIUC, Guest Lecture at the Network Analysis Course April 2024
- ***Information Aggregation and Distributed Learning in Privacy-critical Environments***
 - CMU, Tepper School of Business OM Seminar* September 2024
 - TOC4Fairness Online Seminar Series* March 2024
 - Rutgers Business School, Seminar* November 2023
- ***Leveraging Large Language Models for Collective Decision-Making***
 - LLMs as Research Tools: Applications and Evaluations in HCI Data Work Workshop Invited Talk, ACM Conference on Human-Computer Interaction* May 2024
 - Microsoft Research, Invited Applied Research Talks August 2023
- ***Structural Measures of Resilience for Supply Chains***
 - Columbia University, Theory Seminar November 2023
- ***Resource Allocation in a Financial Contagion Environment***
 - University of Chicago, Theory Seminar October 2023
 - Indiana University, Center for Complex Networks and Systems Research September 2023
 - Cornell University, CS 6850 Guest Lecture April 2023
 - Cornell University, Theory Seminar November 2022
- ***GLINKX: A Scalable Unified Framework for Homophilous and Heterophilous Graphs***
 - Twitter, Machine Learning Seminar August 2022

DEPARTMENTAL
AND
PROFESSIONAL
SERVICE

Department and School Outreach

- AI and Data Analytics Center (AIDA) Brown Bag Seminar Series Co-organizer [\[link\]](#)
- Department of Information Systems Brown Bag Seminar Co-organizer [\[link\]](#)
- W. P. Carey School of Business, PhD Lunch and Learn, November 2025, Speaker [\[link\]](#)

Program Committee

- ACM Conference on Fairness Accountability and Transparency (FAccT)
- ACM Conference on Knowledge Discovery and Data Mining (KDD)
- ACM Conference on Economics and Computation (EC)

Ad-hoc Reviewer

- Management Science
- MIS Quarterly
- INFORMS Journal of Computing
- Mathematics of Operation Research
- ACM Transactions on Management Information Systems
- Journal of Artificial Intelligence Research
- Naval Research Logistics
- International Conference on Computational Social Science (IC2S2)
- ACM Web Conference (WWW)
- Innovations in Theoretical Computer Science (ITCS)
- ACM Conference on Human Factors in Computing (CHI)
- European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases (ECML-PKDD)
- Machine Learning
- NeurIPS 2022 Workshop on Graph Learning Frontiers (GLFrontiers)
- NeurIPS 2021 Workshop on Human and Machine Decisions
- Journal of Open Source Software
- IEEE International Conference on Trust, Privacy and Security in Intelligent Systems, and Applications (TPS)

Session Chair

- | | |
|---|------|
| – INFORMS Annual Meeting | 2025 |
| – NSF Conference on Network Science and Economics | 2024 |
| – ACM Web Conference (WWW) | 2022 |

REFERENCES

Available upon request.